

Smart Tourism Information System

- Village Traveler -

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1.0. Introduction and Background

Wathugedara is a semi-urban area in the Balapitiya Divisional Secretariat Division of the Galle District in the Southern Province. The area acts as a major connecting point between the coastal belt stretching from Bentota to Hikkaduwa and the inland wetland ecosystems of the Madu River. The main reasons for tourists to visit the area are environmental exploration (Madu River Boat Safari), cultural exploration (traditional cinnamon industry, Meetiyaagoda Sandakadapahana Mines) and architectural sites (Jeffrey Bawa's Lunu Ganga and Brief Watta). However, most tourists stay in major hotels and make only short trips, and miss out on important sites such as the colossal reclining Buddha statue at the Karandeniya Shailathalarama Viharaya in the interior villages. I propose this information system to overcome the shortcomings of digital information and maximize the tourism potential of this area.

1.1. Problem Statement

The main problem faced by tourists visiting Balapitiya and Wathugedara is the lack of official and reliable information on essential tourist information. Information on correct routes, GPS locations, and opening hours to boat safaris, museums, and inland cultural sites is not available in one place. Filling this information gap is essential as it will enable tourists to travel independently and confidently in rural areas, thereby boosting the economy of the local people.

1.2. Objective

The main objective of this project is to create a location aware website called "Village Traveler" that will help users explore cultural, environmental and historical attractions within a 25 km radius of the estate and easily plan day trips.

1.3. Project Objectives

To create a cloud-based database (Firebase) containing photos, historical details and GPS coordinates for 10 major tourist attractions around Wathugedara.

To develop an algorithm that calculates the distance from the user's current location to the tourist attraction and displays the closest locations first.

To provide all location details in both Sinhala and English for the convenience of local and foreign tourists.

To integrate the Turn-by-turn navigation feature by connecting directly with Google Maps.

To load the application's information within 3 seconds under a normal 4G network.

1.4. Scope and Limitations

1.4.1. Scope

The application is only available within a radius of 25 km from the city center of Wathugeda. It includes information on 10 tourist attractions, live distance display, and facilities for planning 'One-Day Trips'.

1.4.2. Limitations

This does not include hotel, vehicle or boat booking facilities, payment gateways, and live public transport timetables. This is an informational application only.

1.5. Target Users

- Independent Tourists: People who travel on a budget, alone, and use smartphones to search for information.
- Cultural and Eco Tourists: People who travel with families or in pairs, and want to enjoy history, art, and the environment (Madu Ganga, Bawa Gardens).
- Domestic Weekend Tourists: Sri Lankans who come south from Colombo or other cities for a weekend getaway and are interested in exploring new places (e.g. Karandeniya Temple).

1.6. Stakeholder Identification Plan

Stakeholder	Role	Key Requirement Identified
Stakeholder 1	System Administrator	Secure authentication, administrative management of tourist locations/categories, map data control, and system maintenance (backups).
Stakeholder 2	Local Tourism Authority	Promotion of destinations via structured categorization, interactive route planning to improve visitor flow, and safety/cultural guidance.
Stakeholder 3	Local Business Owner	Platform visibility for business listings (contact, hours, location) and the ability for owners to update their own business information.
Stakeholder 4	Tourist	Intuitive destination exploration, detailed place information (fees/hours), and a customizable one day visit planning tool with save features.
Stakeholder 5	Local Guide	Precise location data for professional use, category-based filtering, and a mobile optimized tool for creating time efficient itineraries on site.

1.7. Overview of the proposed system

As soon as the user navigates to the website, the phone's GPS detects their location and displays the nearest tourist attractions on the screen first. When a relevant place is selected, its history, photos, and travel time are displayed, and when the "Navigate" button is pressed, it is directly redirected to Google Maps. It also includes special "day trip" plans that combine several nearby places.

1.8. Preliminary List of Places

Place Name	Category	Distance from your location	Description of Places
1. Madu Ganga Safari	Nature / Ecotourism	3km	A rich ecosystem famous for its mangroves and high biodiversity.
2. Kothduwa Rajamaha Viharaya	Religious / Historical	4km	An ancient Buddhist temple located on an island within the Madu River.
3. The Ancestral Home of Masks & Kolam Drama	Culture / Arts	5km	A center in Ambalangoda showcasing traditional Sri Lankan mask carving & Kolam drama mask
4. Meetiyagoda Moonstone Mines	Industry / Tourism	7km	The main site in Sri Lanka for mining authentic Blue Moonstones.
5. Karadeniya Shailathalarama Viharaya	Religious / Historical	6km	An ancient temple featuring one of the largest reclining Buddha statues in South Asia.
6. Kosgoda – Cinnamon Training Academy Limited	Industry / Tourism	11km	Sri Lanka main training center about Cinnamon. With Cinnamon Factory & Filed GMP model.
7. Ahungalla Beach	Coastal / Leisure	10km	A quiet and peaceful stretch of beach away from the usual crowds.
8. Hikkaduwa Coral Sanctuary	Marine Life	19km	A marine national park perfect for viewing colorful coral reefs.
9. Lunuganga Estate	Architecture / Garden	22km	The famous country estate and gardens designed by Geoffrey Bawa.
10. Brief Garden	Architecture / Garden	20km	A beautifully landscaped garden created by artist Bevis Bawa.

1.9 Technology Uses

Frontend : React or Angular

Backend : Spring Boot

Database : Mongo DB or MySQL

Development Tools : IntelliJ IDEA

Version Control : Git

Conclusion

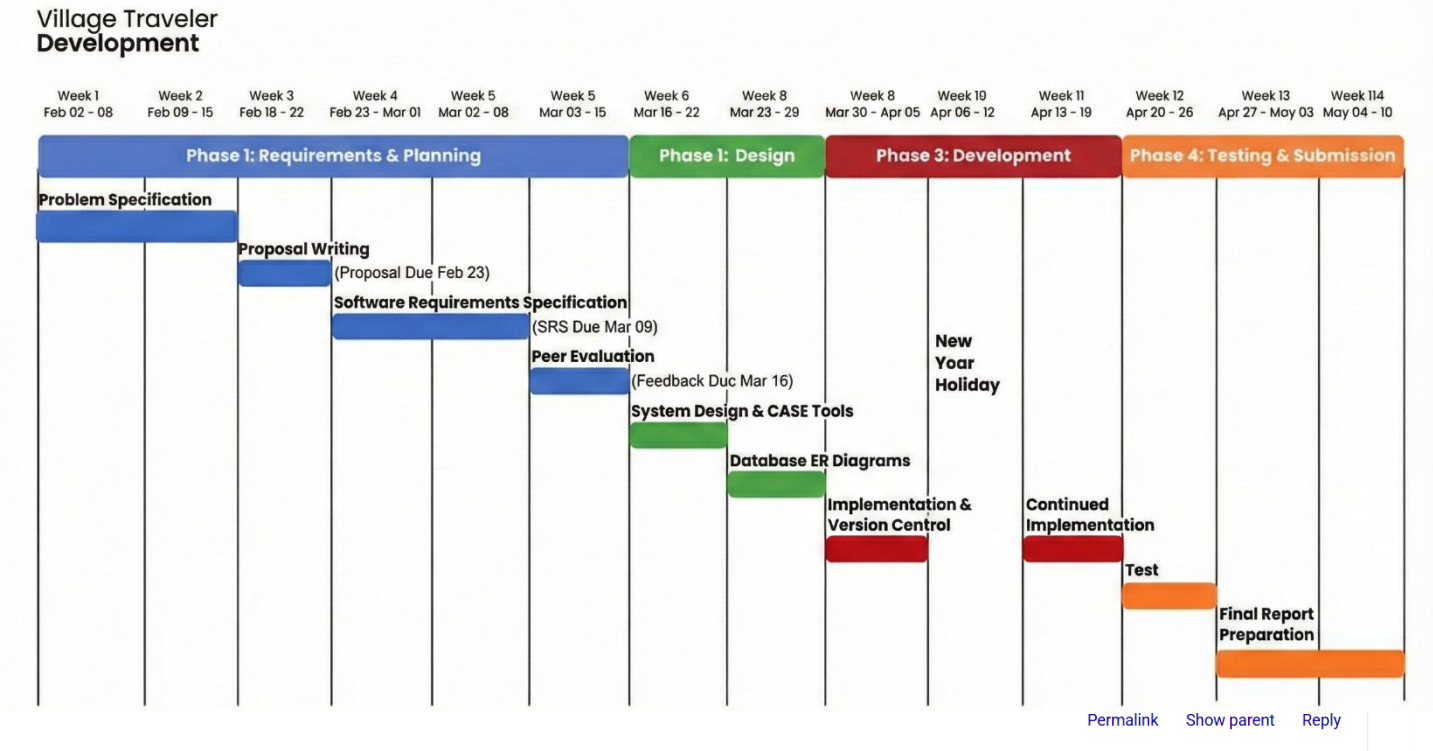
This project is extremely important as it will direct tourists who are limited to the main coastal belt to the inland rural areas around Wathugedara and Balapitiya. This will not only enable tourists to obtain accurate information without any hassle, but will also directly contribute to the upliftment of the local economy and small business owners in areas like Kosgoda, Karandeniya and Meetiyagoda.

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Appendices

Appendix A



Re: Stakeholder Analysis
by [E2320340 Thotawaththa MP](#) - Thursday, 19 February 2026, 6:30 AM

Hi everyone...
I'll be the Restaurant Owner then.

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Re: Stakeholder Analysis
by [E2320340 Thotawaththa MP](#) - Friday, 20 February 2026, 7:51 PM

I'll be the local tourism authority. sorry for my mistake everyone

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Re: Stakeholder Analysis
by [E2320340 Thotawaththa MP](#) - Friday, 20 February 2026, 8:07 PM

From the perspective of a **local tourism authority**, the system should support the promotion of tourism by providing visitors with clear, accurate, and well-organized information about tourist destinations. The platform should present categorized tourist places such as cultural attractions, natural sites, recreational areas, and services in a structured manner, enabling visitors to easily discover and plan visits. An interactive map should be used to visualize locations, show nearby attractions, and support route planning to improve visitor movement and distribution across destinations. The system should also communicate general visitor guidance related to safety, cultural awareness, and responsible tourism to enhance the overall travel experience.

In addition to functional capabilities, the system must meet key non-functional requirements to ensure quality and reliability. The platform should be easy to use, even for first-time visitors, with a simple interface and intuitive navigation. It should be responsive and mobile-friendly so tourists can access information conveniently while traveling. The system should perform efficiently with fast load times and stable operation, minimizing errors or service interruptions. Content presentation should be clear and accessible, supporting a wide range of users and devices, while allowing administrators to maintain and update tourism-related information efficiently.

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Re: Stakeholder Analysis
by [E2320316 Gunasekara VP](#) - Friday, 20 February 2026, 5:09 PM

For System Administrator stakeholder role, I was able to identify that the system must provide secure login functionality and allow management of tourist places, categories, and map data. Additionally, the system must ensure security, reliability, usability, and performance efficiency. Moreover, these can be further discuss as :

Function Requirements - Authentication, Manage tourist places, Manage categories, Manage places information, Map data management, Data control, Backup and Maintenance.

Non Functional Requirements - Security, Performance, Usability, Reliability, Scalability, Compatibility and Maintainability

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